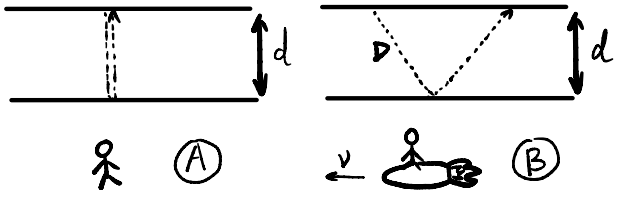
time dilation

consider a photon bouncing,



case A: $T_A = \frac{2d}{c}$

case B: $D = \sqrt{d^2 + (v\frac{T_B}{2})^2}$

$$T_B = \frac{2D}{c}$$

$$T_B^2 = \frac{4\left[d^2 + (v\frac{T_B}{2})^2\right]}{c^2}$$

$$T_B^2 = \frac{4d^2}{c^2} + \frac{v^2}{c^2}T_B^2$$

$$T_B^2\left(1 - \frac{v^2}{c^2}\right) = \frac{4d^2}{c^2}$$

$$\therefore T_B = \frac{\frac{2d}{c}}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{T_A}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$\boxed{t = \gamma\tau} \quad \text{where } \gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

relativistic momentum and mass

$$\underline{p} = m_0 \underline{v} \quad \text{where} \quad \underline{v} = \frac{d\underline{x}}{dt} \quad \leftarrow \text{1-D case}$$

$$\underline{F} = \frac{d\underline{p}}{dt}$$

Suppose $\underline{p} = m_0 \frac{d\underline{x}}{d\tau}$ where τ is the proper time

$$= m_0 \frac{d\underline{x}}{dt} \frac{dt}{d\tau} \quad \leftarrow \quad t = \gamma\tau, \quad dt = \gamma\, d\tau$$

$$= \gamma m_0 \underline{v}$$

$\therefore$ Relativistic mass $\boxed{m = \gamma m_0}$

Derive $E = mc^2$

$$E = Fs = \int F \cdot ds$$

$$F = \frac{dp}{dt} = \frac{d(mv)}{dt} = m\frac{dv}{dt} + v\frac{dm}{dt}$$

$$\Rightarrow E = \int \left(m\frac{dv}{dt} + v\frac{dm}{dt}\right) ds = \int (mv\,dv + v^2 dm)$$

(E)

$$\frac{dm}{dv} = \frac{d(\gamma m_0)}{dv}$$

$$= m_0 \left(-\frac{1}{2}\right)\left(1-\frac{v^2}{c^2}\right)^{-\frac{3}{2}}\left(-\frac{2v}{c^2}\right)$$

$$= m_0 \left(1-\frac{v^2}{c^2}\right)^{-\frac{3}{2}} \frac{v}{c^2}$$

$$= \frac{m_0 v}{c^2} \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} \frac{1}{1-\frac{v^2}{c^2}}$$

$$= \frac{v}{c^2\left(1-\frac{v^2}{c^2}\right)} \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$= \frac{mv}{c^2-v^2}$$

$$c^2 dm - v^2 dm = mv\,dv$$

Substitute into (E)

$$\therefore \quad E = \int (mv\,dv + v^2 dm)$$

$$= \int c^2 dm$$

$$\boxed{E = mc^2}$$